

VTOL, STOL, and Speed

This was a glorious decade for aircraft development, as world optimism for widespread supersonic air travel peaked and experimental aircraft were built to test all the complex issues involved in achieving that safely. Short-take-off-and-landing (STOL) or vertical-takeoff-and-landing (VTOL) fighter jets went from calculation and experimentation to flying reality and a rocket-powered prototype aircraft set an enduring world manned speed record.

▷ **North American X-15 1960**

Origin US
Engine 70,400 lb (31,933 kg) thrust Reaction Motors Thiokol XLR99-RM-2 liquid-fuel rocket
Top speed 4,520 mph (7,274 km/h)

First flown in 1959, this remarkable rocket-powered research aircraft, released from a B-52, reached outer space (more than 62 miles/100 km above the Earth) and still holds the world manned aircraft speed record.



△ **Bell X-14 VTOL Test Bed 1960**

Origin US
Engine 2 x 2,950 lb (1,338 kg) thrust General Electric J85 turbojet
Top speed 186 mph (299 km/h)

First flown in 1957 with Armstrong Siddeley Viper turbojets, the X-14 was transferred to NASA in 1959 for further VTOL research, including moon landing tests, flown by astronaut Neil Armstrong.



◁ **Hawker Siddeley P.1127 1960**

Origin UK
Engine 15,000 lb (6,804 kg) thrust Bristol Siddeley Pegasus 5 vectored-thrust turbofan
Top speed 710 mph (1,142 km/h)

Privately funded development by Bristol Engines and Hawker Siddeley in the late 1950s led to the first flight in 1960 of what would become the Harrier "Jump Jet": the first successful VTOL fighter.



△ **Bristol 188 1962**

Origin UK
Engine 2 x 14,000 lb (6,350 kg) thrust de Havilland Gyron Junior DGJ 10 afterburning turbojet
Top speed 1,345 mph (2,165 km/h)

Conceived in the 1950s for advanced Mach 3 research, the 188 used new materials such as a chromium stainless steel skin and fused-quartz windshield. It never reached its design speed.



△ **Handley Page HP115 1961**

Origin UK
Engine 1,900 lb (862 kg) thrust Bristol Siddeley Viper BSV.9 turbojet
Top speed 248 mph (399 km/h)

Tested successfully over 12 years, this aircraft was part of the Concorde development project to test low speed handling of delta wings. Despite wings with a low aspect ratio (75 degrees), it could fly as slow as 69 mph (111 km/h).



△ **Dassault Balzac V 1962**

Origin France
Engine 4,850 lb (2,200 kg) thrust Bristol Siddeley Orpheus BOR 3 Cruise turbojet, plus 8 x 2,160 lb (980 kg) thrust Rolls-Royce RB108-1A lift turbojet
Top speed 686 mph (1,104 km/h)

Dassault converted a Mirage III fighter for vertical takeoff and landing, with eight lift engines around the main propulsion engine. Only one was built. It flew successfully but had two fatal crashes when hovering and was not repaired after the second incident.